

Md Kamrul Islam

Lecturer in Computer Science | AI Researcher

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RESEARCH PROFILE

AI researcher with a background in deep learning and medical imaging, with recent research experience in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and information retrieval. My current research interests include foundation models, agentic AI, multimodal AI, and frugal AI.

EDUCATION

Erasmus Mundus Joint Master's Degree in Big Data Management and Analytics (BDMA) Sept 2023 – Oct 2025
CentraleSupélec | Universitat Politècnica de Catalunya (UPC) | Université Libre de Bruxelles (ULB) France | Spain | Belgium

- **Thesis:** *A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models*

Bachelor of Engineering in Software Engineering Mar 2018 – Dec 2021
Sichuan University Chengdu, China

- **CGPA:** 3.64/4.00 (Weighted Average: 87%)
- **Thesis:** *Brain Tumor Detection and Classification Using Convolutional Neural Networks*

RESEARCH EXPERIENCE

AI Research Engineer Intern [\[Project\]](#) May 2025 – Oct 2025
LISSI, Université Paris-Est Créteil | in collaboration with CEA-LIST Paris, France

- Developed a hybrid framework combining LLM-based reasoning with rule-based validation for automated security annotation generation in Business Process Model and Notation (BPMN).
- Constructed a benchmark dataset and evaluated multiple LLMs using different prompt settings and RAG.
- Evaluated the framework against domain-expert annotations, achieving 0.58 precision and 0.52 recall while reducing annotation time by approximately 95%.

Methods & Tools: LLMs, RAG, Python, LangChain, Flask, React.js, CI/CD

Graduate Research Assistant [\[Project\]](#) Oct 2024 – Mar 2025
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), CentraleSupélec Paris, France

- Developed a deep-clustering framework combining Convolutional Neural Networks (CNNs) and Group-Equivariant CNNs (G-CNNs) for a large-scale chest X-ray dataset.
- Investigated geometric inductive biases to improve feature representation and reduce reliance on data augmentation.

Methods & Tools: PyTorch, G-CNNs, OpenCV, CUDA, HPC

PUBLICATIONS

1. **M. K. Islam**, T. Henry, M. Salnitri, J. Köpke, and S. Souihi, “A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models,” *24th International Conference on Business Process Management (BPM 2026) – Business Process Management Forum*, Toronto, ON, Canada, pp. 168–184. In *Lecture Notes in Business Information Processing*, vol. 599, Springer, Cham. [\[DOI\]](#)
2. C. K. Sah, L. Xiaoli, M. M. Islam, and **M. K. Islam**, “Navigating the AI Frontier: A Critical Literature Review on Integrating Artificial Intelligence into Software Engineering Education,” *36th International Conference on Software Engineering Education and Training (CSEE&T)*, Würzburg, Germany, 2024, pp. 1–5. [\[DOI\]](#)

ACADEMIC & PROFESSIONAL EXPERIENCE

Lecturer [Faculty Profile] Department of Computer Science, American International University-Bangladesh (AIUB)	April 2026 – Present Dhaka, Bangladesh
- Teach undergraduate courses in Data Processing, Data Visualization, and Programming for Data Science. - Develop course materials, laboratory exercises, and project-based learning activities for undergraduate courses. - Supervise undergraduate research projects in problem formulation, experimentation, and scientific writing.	
Big Data Engineer Intern Chengdu Suncap Co., Ltd.	Dec 2020 – May 2021 Chengdu, China
- Developed an end-to-end data pipeline for large-scale data ingestion, transformation, and analytics in distributed environments. - Enhanced data preprocessing and feature engineering, improving predictive model accuracy by 20%. - Collaborated in an Agile Scrum environment to implement modular software design and CI/CD practices.	
Methods & Tools: Apache Spark, Scikit-learn, SQL, Git, Jira, CI/CD	

TECHNICAL SKILLS

Programming: Python, C/C++ , SQL, JavaScript
AI/ML: PyTorch, Scikit-learn, OpenCV, Hugging Face Transformers, LLM APIs, LangChain
Big Data & Cloud: Apache Spark, Microsoft Azure, Microsoft Fabric, Apache Airflow
Databases: PostgreSQL, Neo4j, GraphDB, SPARQL
Computing: Linux, Docker, CUDA, HPC, Git

PROJECTS

DigiScan360 [Project] Developed a competitive intelligence platform and pitched it as a startup prototype. Tech Stack: PySpark, LLMs, Microsoft Fabric, Microsoft Azure, Power BI, GraphDB, SPARQL	Feb 2024 – June 2024
Anomaly Detection in Diesel Train Cooling Systems [Project] Developed unsupervised models to detect anomalies from spatio-temporal data in train cooling systems. Tech Stack: Python, SQL, Scikit-learn, Tableau	Sept 2023 – Dec 2023
Brain Tumor Detection and Classification Using Convolutional Neural Networks [Project] B.Eng. capstone project: achieved 98% classification accuracy using a custom CNN model for multi-class brain tumor classification. Tech Stack: Python, TensorFlow, Keras, OpenCV	Sept 2021 – Dec 2021

ACADEMIC SERVICE & ACTIVITIES

Peer Reviewer – 4th International Conference on Computing Advancements, 2026 [Certificate] ACM SIGAPP In-Cooperation American International University-Bangladesh	Aug 2026 Dhaka, Bangladesh
Twelfth European Big Data Management & Analytics Summer School (eBISS 2024) [Poster] University of Padova	July 2024 Padova, Italy

HONORS & SCHOLARSHIPS

Erasmus Mundus Scholarship European Commission – Fully funded merit scholarship for the BDMA Joint Master’s Programme	2023 – 2025
Belt and Road Initiative Scholarship Chinese Government – Fully funded scholarship for undergraduate study at Sichuan University	2018 – 2022

LANGUAGES

English: Proficient Chinese: Upper-intermediate French: Elementary Bengali: Native
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